

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

19 September 2026

Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, a winding field and a uniform one are stationary at one coupling with opposite resultants, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together

with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are

21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 8.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand's relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator's stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable

44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
51 composes them, stating the thermodynamic constraints and all eight con-
52 necting hypotheses per branch; its supporting models of feedback, learning
53 and resource supply have distinct state spaces and assumptions. Section 6
54 then examines what the observables can establish, before Section 7 devel-
55 ops the cortical hypothesis and an awakening experiment. The hypothesis is
56 that extracellular contraction increases field-mediated coupling; its direction,
57 rate and relation to recovery require independent calibration. Supplemen-
58 tal Material organizes the constructions and controls behind the claims they
59 support.

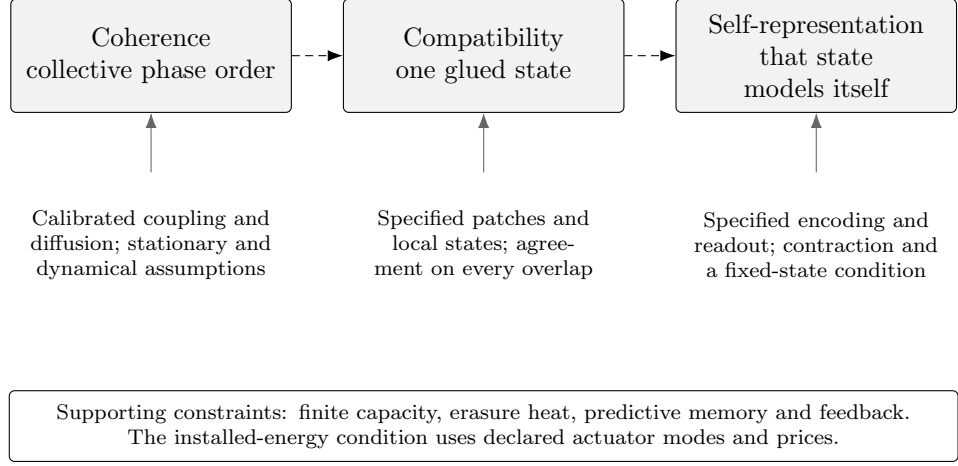


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

2 Coherence: physical coupling and its resource condition

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero.

83 These linear estimates concern the limiting equation. Relating them to finite
84 stochastic trajectories requires a dynamical mean-field limit and branch selec-
85 tion; exchange symmetry alone does not establish the needed independence
86 (Sznitman, 1991).

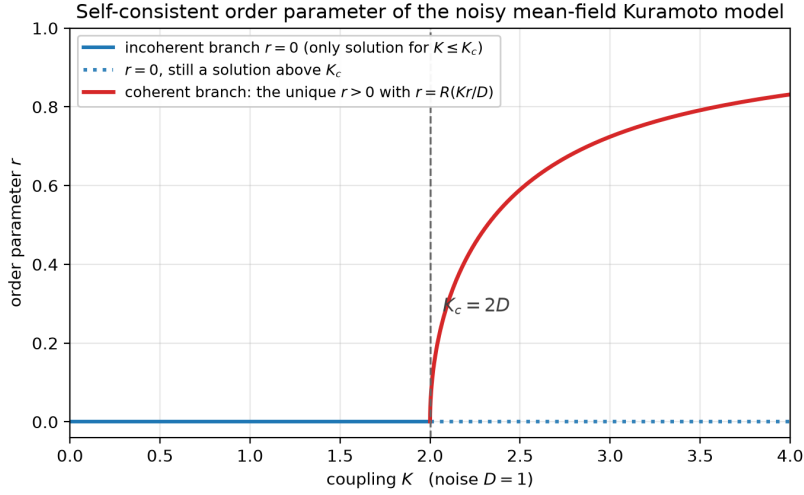


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

87 2.2 Installed energy bounds mean coupling

88 A spatial model replaces the finite sum by integration against a kernel. One
89 declared implementation carries response modes with continuous profiles ϕ_i ,

90 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

91 The same configuration sets the reciprocal kernel and stored energy. An
 92 executed transition changes the kernel through its changed occupancies and
 93 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 94 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 95 pair reconstruction on refining spatial partitions converges to its product-
 96 measure integral under the specified compactness and sampling assumptions;
 97 the refinement index is independent of the number of control updates.

98 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 99 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 100 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

101 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 102 coherent stationary branch and any growing Fourier mode about the uniform
 103 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 104 is necessary: energy above it need not produce sufficient coupling because
 105 the bound can be loose.

106 The quantity constrained is installed energy. Dissipated heat does not
 107 determine it, and κ is supplied by the mode profiles, prices and substrate
 108 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 109 tial dynamics requires a justified reduction. Identifying the installed kernel
 110 with a cortical field is a separate physical commitment.

111 **3 Compatibility: from coordinated activity** 112 **to a global description**

113 **3.1 Why common timing is insufficient**

114 For overlapping patches U and V , compatibility means that restricting either
 115 local description to $U \cap V$ gives the same object. Descriptions can agree on
 116 shared quantities while contributing different information elsewhere. Com-
 117 patible sections of the specified sheaf have a unique global extension. On
 118 a finite cover, local finite spatial measures agreeing as measures on every
 119 overlap also have a unique finite-measure extension whose sheaf image is the
 120 glued section. No mass-one normalization is imposed.

121 In the cup example, visual and reaching descriptions must express a
 122 shared quantity, such as cup location, in common coordinates. Restriction
 123 retains that quantity while dropping details particular to each description.
 124 Agreement permits surface appearance and hand configuration to contribute
 125 to one situation. The mathematical counterexample assigns the same phase

126 to every patch while two local mass profiles disagree on a shared site. Those
 127 profiles have no common extension. Because they are chosen independently
 128 of phase, synchrony places no constraint on their disagreement.

129 This distinction limits the biological interpretation. Receptive territories
 130 and content domains must be related to spatial patches, and the descriptions
 131 must support the required extension. Matching overlap marginals of de-
 132 coded probability laws need not determine a unique global joint distribution
 133 (Abramsky & Brandenburger, 2011).

134 **3.2 When coherence does constrain content**

135 A declared relation between phase and content gives a positive result. Let
 136 contents take values in a real normed vector space E . Patch i reads an
 137 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 138 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 139 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 140 sites; their equality on overlaps is a supplied relation between descriptions of
 141 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 142 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 143 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

144 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 145 agree exactly. The factor N makes this estimate potentially weak in large
 146 populations; it supplies a pointwise guarantee from a population average.
 147 The population in it need not be the whole array. A patch carries an order pa-
 148 rameter of its own, and the same bound read there carries that patch's resul-
 149 tant and that patch's site count (`chord_le_of_patch_coherence, compatible_of_patch_coherence`
 150 so a large population observed through small patches is constrained by the
 151 small count. A shared scalar encoder is a special case. Dropping the overlap
 152 relation permits disagreement even at perfect locking. A common nonex-
 153 pansive linear projection preserves agreement, but agreement of a projection
 154 does not establish agreement of the full vector.

155 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 156 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 157 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 158 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

159 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 160 duce an initial mismatch. Approximate observations leave a nondecaying
 161 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 162 update to the phase process.

163 3.3 What approximate agreement determines

164 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
165 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
166 Weights that vanish outside their patches and sum to one at each site select
167 a global profile by weighted averaging. The selection lies within ε of each
168 patch there, and any two such selections differ by at most ε , independently
169 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
170 $L\varepsilon$ across selections.

171 At zero tolerance, the selection is the unique exact extension. At every
172 positive tolerance there are families with no exact extension; selection then
173 gives an approximation whose dependence on weights is bounded. Supple-
174 mental Material gives a three-site witness attaining the bound. The mass-
175 profile construction uses the scalar case of the content dynamics and addi-
176 tionally requires nonnegativity of the initial profiles and observations. The
177 general vector result supplies no identification with the spatial-measure sheaf.

178 These constructions concern declared objects: a specified cover, contents
179 valued in a normed space, and encoders related on overlaps. They select
180 neither the neural cover nor the decoder. Identifying their contents with
181 perceptual content, and establishing that the identification bears on experi-
182 ence, is a further modelling task; Section 6 sets out what a measurement of
183 it would have to establish.

184 4 Self-representation: a state that models it- 185 self

186 4.1 The reflexive construction

187 Compatibility gives a global state. Reflexivity relates it to an encoding within
188 that state. Equip global sections with a complete metric. If the additional
189 restriction-resonance condition identifies encoding with restriction to a region
190 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

191 The region and readout are fixed across the state space. If F contracts and
192 fixes the section s_* formed from the local descriptions, Banach's theorem
193 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
194 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
195 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
196 one for $\tau > 0$. Constructing a map with that factor remains a modelling
197 obligation.

198 In the cup example, the global state includes the represented scene and the
199 perceiver's relation to it. A local encoding must retain enough information
200 for the readout to recover that modelled state; cup position alone would be
201 insufficient if hand configuration and viewpoint could not be reconstructed.

202 For the same map with Lipschitz factor $q < 1$, approximate reconstruction

203 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1-q}. \quad (9)$$

204 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 205 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 206 weakens.

207 4.2 Reconstruction across a family of states

208 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 209 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 210 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

211 Thus M relevant states pairwise more than 2ε apart require at least M
 212 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 213 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 214 existing self-prediction map.

215 This is an additional accuracy criterion on a family, with no requirement
 216 for multiple exact fixed points. A blind encoder or constant readout fails it
 217 for separated states despite having a unique fixed point.

218 4.3 Agreement after a change takes communication

219 Snapshot compatibility does not specify how quickly descriptions respond
220 when a shared quantity changes. In a synchronous network whose updates
221 read only the site’s own state and declared neighbours’ messages, two initial
222 configurations agreeing on a site’s causal past through T rounds produce the
223 same state there at round T . If two fresh interventions outside that past
224 require reports separated by more than 2ε , no report from that site can be
225 within ε of both. This follows from the reconstruction bound in Section 4.2.

226 A three-site delay line witnesses failure after one round and exact report-
227 ing after two; cutting the path prevents delivery at every round. A report
228 may still happen to be correct about one fixed world. The bound concerns
229 responsiveness across interventions.

230 Throughout this section *prediction* names a reconstruction map; reading
231 it as temporal neural prediction would require dynamics. The relevant family
232 of states, the metric, the tolerance, the communication graph and the dead-
233 line are all declared inputs, as are the represented variables and the fidelity
234 counted as accurate. Reading the resulting code count as physical memory
235 capacity requires a hardware model, and a readout that varies with the state
236 requires a further one.

237 5 Thermodynamics and composition

238 The phase and representation results concern coupling, content disagreement
239 and reconstruction error. Thermodynamics constrains the physical opera-
240 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
241 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-
242 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
243 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
244 composition connects these objects through the explicit hypotheses in Ta-
245 ble 1.

246 5.1 Feedback changes the information constraint

247 For passive memory X_t driven by an environmental signal S_t , the predictive
248 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

249 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
250 et al., 2012). Zero retained information also satisfies it, so a task and a
251 learning rule require additional specification. Relating this bound to Lan-
252 dauer heat requires a common thermal scale and budget, and memory waste
253 bounded by erased entropy.

254 An acting agent changes the signal it will receive. Specify policy, world,

255 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{12}$$

256 The second conditional term vanishes in the passive model. With feedback,
257 the signed difference can be negative as action creates correlations with the
258 future. A causal reading requires an environmental state sufficient to describe
259 the relevant dynamics.

260 For a controlled substep, the active thermodynamic bound instead uses
261 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
262 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
263 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
264 2013). Every forward path of positive probability must have a positive-
265 probability reverse for this finite identity. A missing reverse gives infinite
266 divergence. The supplement develops the support conditions and composes
267 actuation with memory updates on their shared intermediate law.

268 **5.2 Learning and repeated operation need explicit re-** 269 **sources**

270 The finite examples distinguish selecting a feasible policy, relaxing a register
271 toward an encoded reward, and learning an unknown task from observa-
272 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 2.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy E , thermal scale θ , local detailed balance and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

5.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.

Table 1 (continued)

Connection	Status	Required assumption
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

307 lap compatibility and a reflexive boundary specifying encoding and readout.
 308 They conclude `UnifiedSelf`: the glued section is the unique fixed point of
 309 self-prediction. Compatibility and accuracy at that section enter as premises;
 310 uniqueness follows. Identifying encoding with restriction needs the separate
 311 resonance condition of Eq. (8).

312 Joint witnesses combine a register, boundary, refining mesh and small
 313 spatial cover; active witnesses add controlled thermal updates and a priced
 314 installation. They establish satisfiability under their stated identifications.
 315 Mesh convergence follows from refinement data, and the witnesses establish
 316 no common cortical mechanism. Supplemental Material maps the claims to
 317 their Lean declarations and records the counterexamples. The development
 318 declares no additional axioms beyond its libraries' foundational axioms.

319 6 What the observations establish

320 The composition holds under its premises. Whether observation can establish
 321 those premises is a separate question, and the three requirements it connects
 322 concern different objects: phase dynamics, local descriptions and internal
 323 encodings. Their observables must retain these distinctions. The following
 324 controls show which mechanisms remain indistinguishable, which temporal
 325 approximations fail, which states one observable cannot separate, and what
 326 independent content measurements must establish.

327 6.1 Stationary distributions leave the mechanism un- 328 determined

329 Equation (2) gives a concentration–coherence relation with no fitted shape
330 parameter. Estimate r from circular resultant length and a from density
331 shape: inverting R from the measured r would make agreement automatic.
332 Even independently estimated agreement identifies a statistical family shared
333 by different mechanisms.

334 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
335 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
336 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
337 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
338 and diffusion D_0 have that same density. Their stationary distributions co-
339 incide window by window despite different causes.

340 Resolved dynamics supply additional information. In the ideal model with
341 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
342 increment means constrain drift. Separating endogenous coupling from a
343 matched common drive additionally requires a measured drive or controlled
344 perturbation; identifying electromagnetic coupling requires evidence specific
345 to that interaction. Supplemental Material gives the equivalence construction
346 and its numerical check.

347 **6.2 Finite ramps change the observable onset**

348 The square-root law in Eq. (3) describes stationary solutions. A changing
349 coupling requires time for the phase distribution to approach those solutions.
350 In a deterministic mean-field control that crosses the threshold symmetrically,
351 none of the scanned rates meets the declared tracking tolerance: slowing
352 the crossing does not remove the terminal residual under that initialization.
353 A slow control variable therefore supplies no general guarantee of branch
354 tracking at onset. Supplemental Material gives the residual, rate range and
355 admissible rates away from threshold.

356 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
357 at four coupling speeds. Escape is the first of three consecutive saved samples
358 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
359 replicas escape within the fastest window. The three slower ramps have
360 mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents
361 differ from those of the exact stationary branch evaluated on the same grids.
362 At the fastest speed the ensemble mean never reaches the fit window, so its
363 fitted slope does not estimate stationary onset.

364 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
365 makes the apparent exponent depend on ramp speed and fitting window.
366 The specified deviation-based critical ramp speed is not bracketed by this
367 sweep. With fluctuating coupling, the relevant stationary comparison also
368 depends on rate: the mean coupling approximates order in the tested fast
369 regime, while averaging stationary order over coupling works for the tested

slow drives confined to one side of threshold. Slow crossing drives can satisfy
neither approximation. These are finite-horizon observations, with controls
and parameters in Supplemental Material.

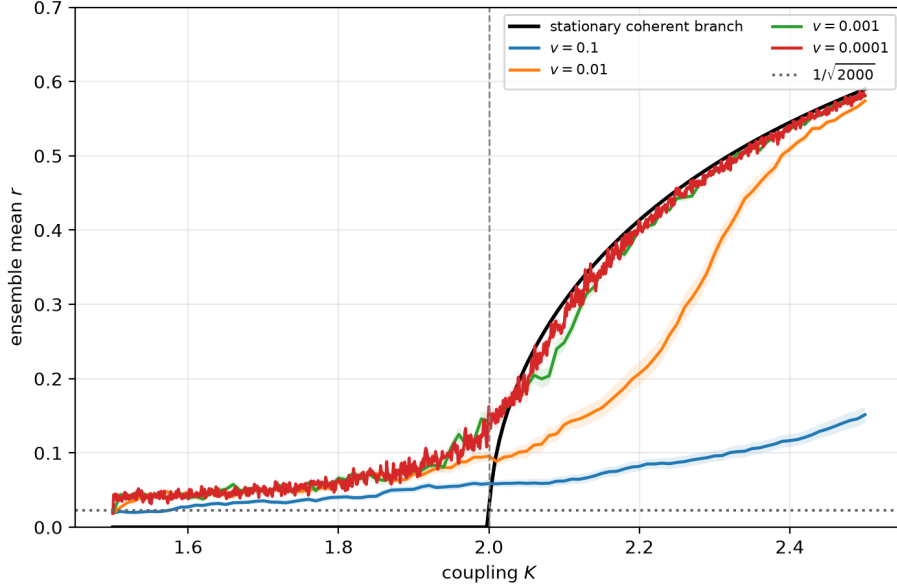


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

Additional controls address integration bias, population size and spatial
or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
rameters and controls in Supplemental Material; they supply no cortical cal-
ibration.

378 6.3 A winding state leaves the order parameter am- 379 biguous

380 The order parameter r is indifferent to where a phase sits in space. A field
381 winding once across a periodic sheet therefore has $r = 0$ while every patch
382 of it stays locked. Such a field is not an artificial construction here. On
383 a torus a twist is stationary for any isotropic symmetric kernel at identical
384 frequencies: the coupling term is odd in the separation against a kernel even
385 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
386 this for any finite periodic array, this sheet included. A twist satisfies the
387 positivity and identical-frequency hypotheses. It violates only the quarter-
388 turn initial spread that the convergence result additionally requires before
389 placing the limit at the uniform state.

390 On the sheet of Section 2's spatial control, a twist survives below an in-
391 terpolated decay length of 0.1871 mm and unwinds above it. That boundary
392 is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same
393 coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and
394 $r = 0.0083$ from a twisted one. Both are stationary, with patch-local order
395 0.9015 and no winding defect. At fixed K the observable therefore takes two
396 values spanning nearly its whole range. Coupling does not determine the
397 resultant, and a low resultant does not by itself distinguish weak coupling
398 from a winding field.

399 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

400 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 401 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realisations.
 402 Heterogeneity shortens the range that sustains a wave without removing it.

403 Which windings persist has a criterion behind it. Linearising the phase
 404 model about a winding returns a coupling of the same translation-invariant
 405 form, each separation reweighted by the cosine of the phase the winding
 406 advances across it (`hasDerivAt_charJacobian`). That reweighted ker-
 407 nel is again even in the separation, so the same characters diagonalise
 408 it and every mode's rate is one of its eigenvalues. Where the wind-
 409 ing advances by at most a quarter turn across every separation the cou-
 410 pling reaches, the reweighted kernel is nonnegative and no rate is positive
 411 (`charJacobian_stable_of_quarter_turn`); where some rate is positive, that
 412 mode grows without bound (`charJacobian_growing_mode`). Moving cou-
 413 pling mass to longer separations is what carries a phase advance past a
 414 quarter turn, and that is the mechanism the decay-length sweep varies. The
 415 criterion bounds rates rather than trajectories: asymptotic stability is not
 416 proved, so the states that persist occupy a narrower range than it admits,
 417 and the boundary quoted above is measured and not derived from it.

418 The same reweighted kernel measures the potential's curvature. A wind-
 419 ing has $r = 0$ where a global minimiser of the Lyapunov potential has $r = 1$,
 420 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 421 tial data does not merely decline to cover a winding, it must exclude it. Along
 422 a mode, the potential's second variation is that kernel's quadratic form, so a

growing mode is a direction of falling potential and a winding carrying one
is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

An amplitude layer decides which of these states exist, and leaves what
the resultant reports unchanged. A phase state carries no amplitude. Every
oscillator sits on the unit circle, so a field with an amplitude node has no rep-
resentation in these dynamics. The real Stuart–Landau extension supplies
one on the same coupling, and leaves intact the gradient structure the con-
vergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\mu - |z_i|^2)z_i +$
 $\sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation with each cou-
pling weighted by a ratio of amplitudes (`amplitude_phase_is_kuramoto`).
The phase model is this one with the amplitude ratios set to one, rather
than a separate object.

What the added degree of freedom decides is which windings exist at all.
On a periodic array of any dimension, this sheet included, a winding requires
 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
summed over separations, with ϕ_d the phase the winding advances across d
(`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
lowers it further. A broad enough kernel therefore removes the winding out-
right. The sweep above reaches that same boundary by a different route:
the phase model retains every winding at every coupling, and loses this one
under integration instead. The two layers bound the same regime from op-
posite sides. A phase recording does not separate a state the amplitude layer

446 excludes from one the dynamics fails to sustain.

447 These states are windings and not spirals, and on the sheet that is a
448 theorem rather than a shortage of dimensions. A cortical phase singularity
449 is a point at which the amplitude vanishes and around which the phase turns
450 once, read in practice as the circulation around a small plaquette. The states
451 above have equal modulus at every site, so that circulation is exactly zero on
452 every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can the
453 layer host a single defect. Over a periodic sheet the circulations of *any* phase
454 field sum to zero, so defects arrive in cancelling pairs (`no_isolated_defect`).
455 The sweep's near-zero measured defect density agrees with the first of these
456 and says nothing beyond it.

457 Two limits bound the layer for a reader of the cortical literature. Relax-
458 ational dynamics has no propagation speed, which is the quantity a cortical
459 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
460 sured envelope is not; this work identifies the two nowhere. Supplemental
461 Material gives the sweep, its uniform control, the amplitude layer and the
462 standing wave it admits inside the band, together with the remaining limits
463 of both.

464 A vanishing resultant is the absence of global agreement and not of
465 agreement. Eq. (6) read on a nearest-neighbour patch of a winding gives
466 a discrepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the
467 patch, which falls to zero as the same winding is spread over more sites
468 (`chord_le_of_char_patch`), while the global form of the same bound is vac-

469 uous at $r = 0$. Encoders shared on overlaps therefore still bound content
 470 disagreement inside a patch of a state that no global observable calls coher-
 471 ent. The winding number itself stays unconnected to content.

472 This qualifies what a phase trace can establish. An onset criterion stated
 473 in r alone registers a change of basin as readily as a change of coupling.
 474 The protocol of Section 7.3 therefore needs a spatial observable beside the
 475 resultant.

476 **6.4 Coherent adaptation can lose environmental align-** 477 **ment**

478 An adaptive coupling rule might supply representational structure, but its
 479 objective must reward the relevant agreement. For symmetric coupling, pair-
 480 wise sine terms cancel in the population mean drift. Minimizing squared drift
 481 at that fixed mean therefore rewards equalizing drift across nodes. In the
 482 frequency-cluster control, this favors coupling between unlike-frequency clus-
 483 ters, whereas the diagnostic environmental template rewards coupling within
 484 clusters.

485 The formalization proves descent of the squared-drift objective for a spec-
 486 ified gradient flow at fixed phases. Its functional differs from thermodynamic
 487 entropy production: at a coherent stationary density the probability current
 488 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
 489 rial gives the operator, its gradient and the conditions of the descent result.

490 The finite control co-evolves noisy phases and symmetric, nonnegative
 491 couplings at fixed total resource. Its template is a diagnostic covariance of
 492 latent signals, never a training input. Across three seeds, second-half order
 493 remains between 0.9167 and 0.9222, while the within-cluster to between-
 494 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
 495 Random-update, relabelled-gradient and frozen-kernel controls are reported
 496 in Supplemental Material.

497 The immediate post-update objective reduction disappears when aver-
 498 aged over complete update intervals. A prespecified grid of 6 rate–cadence
 499 combinations contains no case meeting the 5% sustained-reduction criterion
 500 while retaining coherence and losing alignment. The finding is coherent adap-
 501 tation away from the specified structure, with no demonstrated sustained cost
 502 reduction.

503 Kernel alignment and overlap compatibility are different observables.
 504 This control addresses acquisition of its designated template under its par-
 505 ticular objective. Learning models can explicitly include data in an objective
 506 (Corberi et al., 2026); an account of agreement about the cup must specify
 507 the content its learning rule is meant to preserve.

508 **6.5 Content agreement needs independent decoders**

509 A neural test needs independently identified domains, local descriptions and
 510 restriction maps. Two decoders should estimate a shared content on overlap-
 511 ping domains before their agreement is examined. Discrepancy requires a null

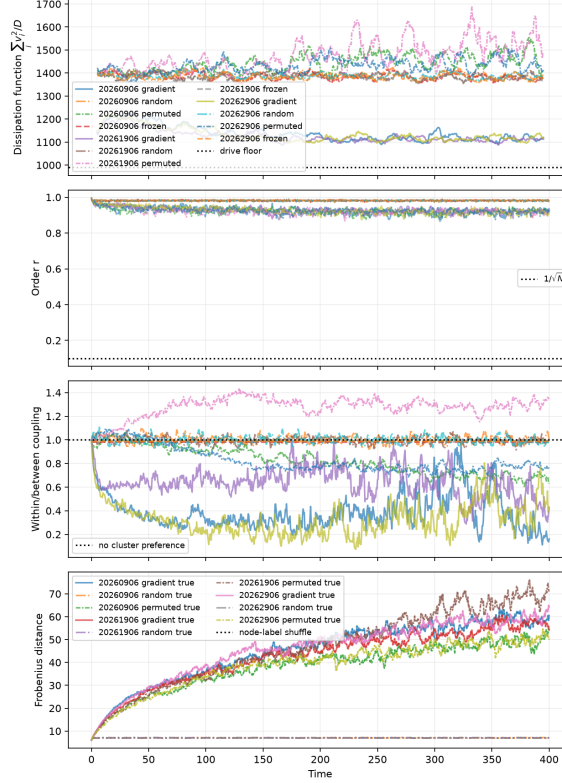


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

512 under decoding noise and controls using deliberately compatible and incom-
513 patible examples. An implementation on constructed data separates these
514 cases at $AUC = 0.990$. Shared-prior shrinkage can nevertheless manufacture
515 agreement, and regional bias can manufacture disagreement. Per-patch de-
516 coding diagnostics detect these failures; the compatibility statistic alone does
517 not.

518 Phase agreement cannot replace that measurement. On constructed con-
519 tent, phase carries a share 1.000 of the content entropy when content is built
520 from it and 0.000 when it is not. A phase-derived statistic reports com-
521 patibility on phase-locked patches whose local descriptions disagree. The
522 implemented content statistic detects their incompatibility. These controls
523 validate a measurement on known constructions; a neural content model and
524 decoder remain to be established.

525 Choosing the cover also chooses a candidate subject: trivial descriptions
526 can agree while carrying little content, and a partition can conceal disagree-
527 ment. The useful choice must therefore be constrained by what neural ac-
528 tivity encodes. The exact composition assumes compatibility of its supplied
529 cover. The content mechanisms above identify additional hypotheses under
530 which agreement can arise; they leave the empirical choice of content and
531 the connection to that cover explicit.

532 6.6 Phase-density estimation and measurement feasi- 533 bility

534 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
535 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
536 itages produce different residuals. The bipolar results are compatible with the
537 Bessel relation, but their concentration range cannot distinguish it from its
538 linear approximation at the observed scatter. Temporal pooling changes the
539 estimand, and sedation recordings leave the awakening hypothesis untested.

540 The concentration range and dependence determine whether the relation
541 can be tested. Under the specified log-density estimator and bias-floor crite-
542 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
543 3000 independent sites in the scanned designs, rising to 31000 when sites
544 share a mean in groups of 10. These counts describe that estimator and de-
545 pendence model; the intended recording requires its own calibration. Supple-
546 mental Material reports the bin-count controls and the full dependence-aware
547 design.

548 An informative dataset needs sufficient concentration range or precision,
549 calibrated estimators and an observation model for dependence, source mix-
550 ing and reference choice. Public mouse calcium imaging provides thousands
551 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
552 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
553 here. Site count, state coverage and bandwidth are separate requirements.

554 **7 The cortical hypothesis and an awakening** 555 **test**

556 The cortical proposal identifies a physical source of coupling and asks whether
557 its changes help explain recovery. The tests in Section 6 determine what
558 observations must distinguish before a phase trace can support this identifi-
559 cation.

560 **7.1 Physical identification and calibration**

561 The cortical hypothesis identifies part of effective coupling with endogenous
562 extracellular electromagnetic interactions. Fields can modulate spike timing
563 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
564 propose coordination of distributed neural information (Pinotsis & Miller,
565 2022; McFadden, 2020). Phase organization under propofol motivates study-
566 ing this regime (Bastos et al., 2021), but does not measure coupling.

567 The formal field predicate requires a jointly continuous kernel, a normal-
568 ized substrate, coupling equal to the kernel average, and no single site pro-
569 viding more than half that strength. Toy constructions satisfy it. A cortical
570 realization must specify the scale at which the kernel and its reduction ap-
571 ply: calcium imaging reports different state-dependent modularity patterns
572 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

573 The spatial reduction is also testable. A row-sum aggregation at zero
574 diagonal tracks the uniform-coupling control over the tested spatial decay

575 range, while frequency heterogeneity displaces the operational threshold. The
 576 fit window biases both estimates relative to $2D$; agreement with the control
 577 is therefore the supported conclusion. Supplemental Material reports the
 578 reduction, its numerical error and the distinct assumptions needed for the
 579 kernel-average identification in Eq. (5).

580 The field commitment requires physical evidence beyond a choice of math-
 581 ematical representation. A finite matrix spread over positive-mass cells pre-
 582 serves its total coupling and phase correlation exactly. Point-supported
 583 vanishing under an atomless measure and a disadvantage from excluding
 584 a better-correlated pair concern specified support constraints. Neither com-
 585 parison supplies energy prices or a biological–digital distinction.

586 A spike-time shift alone supplies no coupling rate. The dimensionless
 587 product of population size N , timing sensitivity s , field amplitude E and
 588 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

589 This rate describes how induced timing shifts accumulate into drift. Nei-
 590 ther it nor the installation factor κ follows from a heat bound. Testing the
 591 physical threshold requires independent estimates of the interaction kernel,
 592 its aggregation and diffusion in the same cortical regime.

593 7.2 Extracellular geometry and recovery

594 The proposed control variable is extracellular geometry. Extracellular vol-
595 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
596 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
597 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
598 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
599 an effective-medium description predicts that a larger extracellular fraction
600 raises conductivity and reduces field amplitude.

601 The coupling-gated hypothesis consequently predicts that extracellular
602 contraction on awakening raises E and, holding the other factors in Eq. (15)
603 fixed, raises effective coupling. This argument supplies a direction; the
604 geometry–coupling relation, its magnitude and its role in human recovery
605 require measurement.

606 For a smooth crossing at nonzero speed, evaluating the stationary branch
607 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

608 This follows from Eq. (3) at fixed D and concerns the stationary branch
609 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
610 dently justified tracking approximation. The crossing controls in Section 6.2
611 leave that approximation unsupported at onset, even though a separation of
612 geometry and phase time scales can support it away from threshold. The

awakening hypothesis therefore needs a calibrated dynamical prediction for an operational onset above the finite-size fluctuation floor. Crossing K_c alone does not supply a recovery latency.

7.3 A protocol for the awakening experiment

The physical test has six requirements:

1. **Coupling and diffusion in the same units.** Estimate them from resolved phase dynamics, separating heterogeneity and observation noise. Measure the drive or perturb the system to distinguish endogenous coupling.
2. **Geometry calibrated to coupling.** Validate the geometry observable and its relation to effective coupling, with uncertainty. The conductivity argument supplies only the expected direction.
3. **Calibrated phase measurements.** Estimate r by resultant length and a by unweighted regression of log density on centered cosine. Choose site count and binning from a dependence-aware calibration over the intended concentration range. The tested designs require thousands of sites at the observed concentrations; temporal pooling changes the spatial estimand.
4. **An independent recovery endpoint.** Measure behavioral recovery separately from phase organization.

633 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
634 old above the finite-size floor and evaluate its latency under the mea-
635 sured coupling trajectory, including ramp speed or fluctuations. Record
636 a spatial observable beside the resultant, since a winding field reports
637 a low resultant at coupling that sustains local order (Section 6.3).

638 **6. Matched model comparison.** Compare coupling increase, diffusion
639 change, common drive and alternative recovery models under the same
640 observation and filtering assumptions.

641 Failure of the concentration–coherence relation over an informative range
642 constrains the density model. Failure of the calibrated geometry–coupling
643 association constrains the physical gate. Better prediction of independent
644 recovery by a competing mechanism counts against its proposed role. A
645 favorable outcome still leaves the content of coordinated activity to be ex-
646 plained.

647 **8 Discussion**

648 Stated positively, and granting the premises collected in Table 1, the account
649 runs as follows. Extracellular contraction on awakening raises endogenous
650 field amplitude and with it the effective coupling between cortical oscillators.
651 Once that coupling passes $2D$, a coherent phase branch exists and the popu-
652 lation settles onto it. Encoders shared where cortical territories overlap turn
653 that phase order into agreement about the quantities the territories describe

654 in common, so the local descriptions glue: there is one state of which each
655 is a restriction. Part of that state encodes the state itself, and a readout of
656 that part reconstructs it accurately enough, and sensitively enough across
657 the states that matter, to make it the unique state the system models itself
658 as being. On this account the glued state is what makes distributed activity
659 one situation, and its accurate self-reconstruction is what gives that situation
660 a perspective.

661 Every step of that description is conditional, and its conditions are the
662 subject of what follows.

663 **8.1 Unity and a minimal self as empirical commit-** 664 **ments**

665 We propose the state glued from compatible local descriptions as a corre-
666 late of conscious unity, and its accurate internal reconstruction as a correlate
667 of a minimal self. The motivations are that different contents belong to
668 one experienced situation and that the situation is experienced from a per-
669 spective. The conditional results identify relations a realization would need:
670 sufficient installed resources for the specified phase model, agreement about
671 shared content, and an encoding that supports reconstruction at the required
672 fidelity.

673 Compatibility allows different descriptions to contribute different infor-
674 mation. It also permits uncertainty and conflicting judgments: apparent cup

675 position and judged cup position can be distinct quantities in a consistent
676 description of their discrepancy. Which quantities are shared must be de-
677 termined independently. Choosing them solely to secure agreement would
678 deprive the proposal of empirical force.

679 The minimal-self proposal concerns the perceiver’s position and condition
680 within the situation, such as the cup’s location relative to a body that can
681 reach it. The encoding could be implicit and need carry no autobiograph-
682 ical narrative. Its proposed necessity is philosophical: even an unreflective
683 episode has a perspective. Gluing by itself requires no self-encoding, and re-
684 constructability alone supplies no functional role. Altering the encoded hand
685 or viewpoint while holding the external scene fixed should change estimates
686 of the perceiver’s relation to the cup as the readout predicts. Whether that
687 represented perspective is experienced remains a further question.

688 **8.2 What a test of the full proposal must establish**

689 Both uniqueness results concern specified objects. They choose neither the
690 cover nor the encoding, and do not select between candidate subjects. In-
691 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
692 here selection requires independent evidence about neural organization. A
693 constant readout can contract and fix a predetermined state while discard-
694 ing its input. The family criterion in Section 4.2 tests the sensitivity that
695 this construction lacks, provided the relevant states and fidelity are declared
696 independently.

697 A test must separately measure coupling relative to diffusion, content
698 compatibility and self-reconstruction, alongside an independent endpoint for
699 experience. Meeting all three without a unified episode would challenge suffi-
700 ciency. A unified episode with incompatible descriptions would challenge the
701 unity identification; one lacking the specified self-encoding would challenge
702 the reflexive requirement. Awakening, dreaming and dissociative states offer
703 possible settings, but absence of report cannot by itself define absence of
704 experience.

705 The next physical requirement is a calibrated geometry–coupling relation
706 and resolved phase dynamics. The installation bound contributes a nec-
707 essary resource condition where its hardware representation applies; cortical
708 mode prices and field-mediated coupling require measurement. For the repre-
709 sentational connection, an informative experiment asks whether a controlled
710 change in coordination alters independently decoded content agreement, with
711 controls for sensory information and decoder quality. The constructed-data
712 estimator establishes feasibility for specified contents. Neural local descrip-
713 tions, shared overlap encoders and the self-encoding have yet to be identified.
714 The population-size dependence of the phase-to-content bound further limits
715 its quantitative use.

716 The separation results explain why these measurements must be inde-
717 pendent. A stationary distribution leaves several recovery mechanisms indis-
718 tinguishable. Equal phase order can accompany incompatible descriptions
719 or different task information. The tested plasticity rule preserves coherence

720 while losing its designated template. Each finding concerns the specified
721 model and observable; their role is to delimit the evidence needed for the
722 proposed connections.

723 **8.3 GPUs and the limits of no-go claims**

724 For a large language model (LLM) running on graphics processing units
725 (GPUs), the framework requires identifying a physical system whose local
726 descriptions form one state and whose internal encoding reconstructs that
727 same state. The coupling-gated route E56–E78 requires a calibrated physi-
728 cal coupling, its noise regime and a connection from coordination to content
729 agreement. An attention matrix or a shared clock supplies none of these iden-
730 tifications on its own. Computing a field model likewise leaves its realization
731 in the computing substrate to be established. These physical obligations also
732 apply to the cortical proposal.

733 Unity requires the candidate’s local descriptions to agree on indepen-
734 dently identified shared quantities (Section 3). Common timing and fluent
735 output do not establish that agreement. Digital representations can sat-
736 isfy compatibility, however, so finite hardware is compatible with the gluing
737 construction. The empirical commitment is to identify the relevant cover
738 and contents with one experienced situation; mathematical uniqueness alone
739 selects no subject.

740 For self, the system must carry an encoding $E(s)$ and an internal readout
741 R of the same global state. E89 requires $F = R \circ E$ to contract and fix

742 the state produced by gluing. A constant readout shows why a unique fixed
743 point alone is insufficient for informative self-representation. The additional
744 family criterion in Section 4.2 supplies a test: if two relevant macrostates
745 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
746 within error ε . An intervention outside the encoding region’s causal past
747 gives a corresponding obstruction before a declared communication deadline.
748 Identical verbal reports alone do not establish identical internal codes. These
749 conditional no-go results exclude a specified reconstruction at a specified
750 fidelity or deadline. They apply to biological and digital candidates alike.

751 The hardware comparisons in Supplemental Material have similarly re-
752 stricted scope. Fixed wiring can impose a correlation disadvantage, and
753 point-supported kernels vanish under an atomless continuum measure. Nei-
754 ther comparison makes the representation conditions unattainable on GPUs.
755 A universal exclusion would need a necessary condition for consciousness and
756 a proof that every relevant implementation violates it. The present condi-
757 tional composition supplies neither.

758 Whether experience requires a living substrate is the explicit subject of
759 biological naturalism (Seth, 2026a,b). On that view the properties ground-
760 ing experience are those of self-sustaining organisms, and simulating them
761 does not secure them. The conditions stated here name no biological mecha-
762 nism. Computation described as such does not satisfy them either: a candi-
763 date must exhibit a physically realized coupling in a measured noise regime,
764 agreement about independently identified shared quantities, and an internal

765 reconstruction meeting a declared fidelity. Whether a realization of those
766 conditions that is not alive would be accompanied by experience is the point
767 at issue, and these results do not settle it.

768 The neuroscientific case against *human-like* consciousness in text-only
769 LLMs concerns embodied sensory interaction, thalamocortical organization
770 and biological regulation linking perception and action to survival (Aru et
771 al., 2023). Their absence supports skepticism, conditional on their causal
772 relevance to experience. Those mechanisms are not requirements of the ab-
773 stract gluing and reconstruction definitions. Workspace-like representations
774 reported in some LLMs (Gurnee et al., 2026) provide candidate organization
775 to investigate. Within this framework, the relevant evidence would identify
776 compatible contents and an informative internal reconstruction, or demon-
777 strate their failure under controlled interventions.

778 8.4 Explanatory scope

779 Integrated-information, global-workspace and predictive-processing accounts
780 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
781 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
782 candidate physical interaction and conditional relations between local content
783 and self-encoding. Gluing and reconstruction permit multiple substrates;
784 the electromagnetic commitment enters through physical calibration. The
785 formalization checks implications under their hypotheses, and its witnesses
786 establish realizability on declared models. They do not together supply a

787 cortical agent.

788 The explanatory question is whether the specified relations occur in neural
789 activity and matter for experience. In the cup example, this means establish-
790 ing how coordination helps maintain one represented situation and how the
791 perceiver’s place within it contributes to the episode. The proposal identi-
792 fies candidate correlates and tests of that identification. Why those relations
793 would be accompanied by experience remains unexplained.

794 **Data and code availability**

795 The Lean development, simulation scripts, tests and saved numerical sum-
796 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
797 `cortical_coherence`. Publication simulation values are generated from
798 saved summaries without rerunning production sweeps. The exploratory
799 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
800 processing and selection are described in Supplemental Material. The repos-
801 itory contains the reproducible analysis and does not redistribute the raw
802 recordings.

803 **Funding and competing interests**

804 This research received no external funding. The author declares no compet-
805 ing interests.

806 Use of AI tools

807 AI assistance to this work was substantial and is not separable by task. Sev-
808 eral coding assistants and chat interfaces were used in an interleaved way
809 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
810 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
811 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
812 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
813 accessed with the Hermes CLI. Between them they contributed to the Lean
814 development, the simulation and analysis code, and the drafting and revision
815 of both documents. Because the tools were used in combination and in alter-
816 nation, no section, proof, figure or result is attributable to any one of them,
817 and none is claimed to be. The author is responsible for all content, including
818 everything a tool produced. Machine checking establishes the stated Lean
819 results under their hypotheses; it does not validate biological interpretations
820 or replace scientific review of generated code and prose.

821 References

- 822 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
823 electric fields emerge from and allow representational drift. *NeuroImage*,
824 253, 119058.
- 825 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-

826 tion theory: from consciousness to its physical substrate. *Nature Reviews*
827 *Neuroscience*, 17(7), 450-461.

828 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
829 Mind. *Oxford University Press*.

830 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
831 tational theory of conscious processing. *Current Opinion in Neurobiology*,
832 25, 76-84.

833 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
834 *Reviews Neuroscience*, 11(2), 127-138.

835 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
836 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
837 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.

838 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
839 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.

840 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
841 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
842 doi:10.1017/S0140525X26106906.

843 Gurnee, W., et al. (2026). Verbalizable Representations Form a
844 Global Workspace in Language Models. *Transformer Circuits Thread*.
845 arXiv:2607.15495.

846 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
847 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
848 40, 24-50.

849 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
850 parts of the free energy principle. *Entropy*, 23(3), 293.

851 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
852 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

853 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
854 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

855 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
856 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
857 *Progress in Neurobiology*, 226, 102465.

858 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
859 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
860 9(8), 45-60.

861 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
862 testable hypothesis about the characteristics of conscious as opposed to
863 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

864 McFadden, J. (2020). Integrating information in the brain’s EM field: the
865 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
866 niaa016.

- 867 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
868 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 869 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
870 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
871 95-103.
- 872 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
873 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 874 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
875 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 876 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-
877 logical Reviews*, 98(1), 239-389.
- 878 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
879 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
880 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 881 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
882 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
883 ity of complexity measures to capture dreaming during propofol sedation.
884 *Scientific Reports*, 15, 32746.
- 885 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
886 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
887 brain. *Science*, 342(6156), 373-377.

888 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
889 *Physiological Reviews*, 88(4), 1277-1340.

890 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
891 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

892 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
893 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
894 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

895 Landauer, R. (1961). Irreversibility and heat generation in the computing
896 process. *IBM Journal of Research and Development*, 5(3), 183-191.

897 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
898 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

899 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
900 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

901 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
902 of prediction. *Physical Review Letters*, 109(12), 120604.

903 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
904 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

905 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
906 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
907 031015.

- 908 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
 909 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
 910 PhysRevLett.111.180603.
- 911 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
 912 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 913 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
 914 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
 915 181.
- 916 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
 917 archive.pitt.edu/115.
- 918 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
 919 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
 920 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 921 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
 922 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
 923 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 924 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
 925 bifurcations with additive noise. *Probability Theory and Related Fields*,
 926 122, 341–388. doi:10.1007/s004400100174.
- 927 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure

928 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
 929 doi:10.1088/1367-2630/13/11/113036.

930 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
 931 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
 932 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
 933 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

934 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
 935 terns underlie complex brain dynamics and are related to cognitive pro-
 936 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
 937 023-01626-5.

938 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
 939 and dissipation with partially observed fluctuating signals. *Physical Review*
 940 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

941 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
 942 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
 943 cell resolution functional networks during unconsciousness are segre-
 944 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
 945 doi:10.1016/j.celrep.2025.116902.

946 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
 947 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
 948 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.